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Online Appendix

Nested Cross-Validated Double Machine Learning for Instrumental Variables in High-Dimensional Settings

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A. NOTATION

For each sample size n, let $\{O_i\}_{i=1}^n$ be a collection of independent observations on a common probability space $(\Omega, \mathcal{A}, P)$. Throughout the main text, I work with i.i.d. sampling; the appendix proofs can be read under i.i.d. sampling with distribution P, or more generally under a triangular array $\{O_{i,n}\}_{i=1}^n$ where within each n the observations are i.i.d. with law $P = P_n$. Each observation is

$$O_i = (Y_i, D_i, Z_i, W_i),$$

where $Y_i \in \mathbb{R}$, $D_i \in \mathbb{R}$, $Z_i \in \mathbb{R}^{d_z}$ with fixed $d_z \geq 1$, and $W_i \in \mathbb{R}^p$ (possibly high-dimensional).

$E[\cdot]$ is written for expectation under P (or P_n when using the triangular-array interpretation).

For a measurable function $f : \mathcal{O} \to \mathbb{R}^q$ with $E[\|f(O)\|_2] < \infty$, define

$$\mathbb{E}_n[f] := \frac{1}{n}\sum_{i=1}^n f(O_i), \qquad \mathbb{G}_n f := \sqrt{n}\Big(\mathbb{E}_n[f] - E[f]\Big).$$

For a vector $a \in \mathbb{R}^q$, $\|a\|_2$ is the Euclidean norm. For a (possibly vector-valued) random variable $f(O) \in \mathbb{R}^q$, define the $L_2(P)$ norm

$$\|f\|_{P,2} := \Big(E[\|f(O)\|_2^2]\Big)^{1/2}.$$

When f is scalar-valued this reduces to $\|f\|_{P,2} = (E[f(O)^2])^{1/2}$. I employ $a \lesssim b$ to mean $a \leq Cb$ for a universal constant $C > 0$ that does not depend on n.

Recall the orthogonal PLIV score (vector-valued when $d_z > 1$):

$$\psi(O; \tau, \eta) := (Z - \pi(W))\Big((Y - \mu(W)) - \tau(D - r(W))\Big) \in \mathbb{R}^{d_z}, \qquad \eta = (\pi, \mu, r).$$

Define the population moment map $M(\tau, \eta) := E[\psi(O; \tau, \eta)] \in \mathbb{R}^{d_z}$.

Let $\eta_0 = (\pi_0, \mu_0, r_0)$ denote the true nuisance functions:

$$\pi_0(W) = E[Z \mid W], \qquad \mu_0(W) = E[Y \mid W], \qquad r_0(W) = E[D \mid W].$$

Define oracle residuals

$$\tilde{Z} := Z - \pi_0(W), \qquad \tilde{Y} := Y - \mu_0(W), \qquad \tilde{D} := D - r_0(W).$$

I also use $g_{ZD,0} := E[\tilde{Z}\tilde{D}] \in \mathbb{R}^{d_z}$ and $\Omega_0 := E[\tilde{Z}\tilde{Z}'] \in \mathbb{R}^{d_z \times d_z}$.

Let $\{\mathcal{I}_k\}_{k=1}^K$ be a partition of $\{1, \ldots, n\}$ into K outer folds, with $\mathcal{I}_{-k}$ the complement of $\mathcal{I}_k$. Let $k(i)$ denote the (unique) fold index such that $i \in \mathcal{I}_{k(i)}$. For each k, the nuisance estimators $\hat{\eta}^{(-k)} = (\hat{\pi}^{(-k)}, \hat{\mu}^{(-k)}, \hat{r}^{(-k)})$ are computed using only the training sample $\mathcal{I}_{-k}$ (including any inner/nested CV).

Define cross-fitted residuals for each observation i:

$$\tilde{Z}_i := Z_i - \hat{\pi}^{(-k(i))}(W_i), \qquad \tilde{Y}_i := Y_i - \hat{\mu}^{(-k(i))}(W_i), \qquad \tilde{D}_i := D_i - \hat{r}^{(-k(i))}(W_i).$$

In the proofs below I implicitly use standard moment conditions ensuring LLN/CLT for the (oracle) score and for residual second moments; e.g. $E[\|Z\|_2^2] < \infty$, $E[Y^2] < \infty$, $E[D^2] < \infty$, and $E[\|\psi(O;\tau_0,\eta_0)\|_2^2] < \infty$.

B. PROOFS

B.1. Proof of Theorem 1 (Identification)

Recall the PLIV model

$$Y = \tau_0 D + g_0(W) + \zeta, \qquad E[\zeta \mid Z, W] = 0.$$

Let $\mu_0(W) = E[Y \mid W]$, $r_0(W) = E[D \mid W]$, and $\pi_0(W) = E[Z \mid W]$, and define the oracle residuals

$$\tilde{Y} := Y - \mu_0(W), \qquad \tilde{D} := D - r_0(W), \qquad \tilde{Z} := Z - \pi_0(W).$$

Step 1: By iterated expectations,

$$E[\zeta \mid W] = E[E[\zeta \mid Z, W] \mid W] = 0.$$

Taking conditional expectations of the structural equation given W yields

$$\mu_0(W) = E[Y \mid W] = \tau_0 E[D \mid W] + g_0(W) + E[\zeta \mid W] = \tau_0 r_0(W) + g_0(W).$$

Subtracting $\mu_0(W)$ from Y and $r_0(W)$ from D gives

$$\tilde{Y} = Y - \mu_0(W) = \tau_0(D - r_0(W)) + \zeta = \tau_0 \tilde{D} + \zeta.$$

Step 2: For any $\tau \in \mathbb{R}$,

$$M(\tau, \eta_0) = E[\psi(O;\tau,\eta_0)] = E\Big[\tilde{Z}\big(\tilde{Y} - \tau\tilde{D}\big)\Big] = E\Big[\tilde{Z}\big((\tau_0 - \tau)\tilde{D} + \zeta\big)\Big] = (\tau_0 - \tau)\, E[\tilde{Z}\tilde{D}] + E[\tilde{Z}\zeta].$$

Since $\tilde{Z}$ is measurable with respect to (Z, W), Assumption 1 implies

$$E[\tilde{Z}\zeta] = E\Big[\tilde{Z}\, E[\zeta \mid Z, W]\Big] = 0.$$

Therefore

$$M(\tau, \eta_0) = (\tau_0 - \tau)\, g_{ZD,0}, \qquad g_{ZD,0} := E[\tilde{Z}\tilde{D}] \in \mathbb{R}^{d_z}.$$

Step 3: By Assumption 2, $g_{ZD,0} \neq 0$. Hence the vector equation $M(\tau, \eta_0) = 0$ holds if and only if $\tau_0 - \tau = 0$, i.e. $\tau = \tau_0$. This establishes unique identification.

B.2. Proof of Lemma 1 (First-order (Gateaux) orthogonality)

Let $O = (Y, D, Z, W)$ be a generic observation and recall

$$\psi(O;\tau,\eta) = (Z - \pi(W))\Big((Y - \mu(W)) - \tau(D - r(W))\Big) \in \mathbb{R}^{d_z}, \qquad M(\tau,\eta) := E[\psi(O;\tau,\eta)].$$

Fix $\tau = \tau_0$. For a direction $h = (h_\pi, h_\mu, h_r)$, consider the path

$$\eta_\varepsilon = \eta_0 + \varepsilon h = \big(\pi_0 + \varepsilon h_\pi,\ \mu_0 + \varepsilon h_\mu,\ r_0 + \varepsilon h_r\big).$$

Define the oracle residual term

$$U_0(O) := (Y - \mu_0(W)) - \tau_0(D - r_0(W)).$$

Step 1: Because the score is bilinear in $(Z - \pi(W))$ and $\big((Y - \mu(W)) - \tau_0(D - r(W))\big)$, I can expand exactly:

$$\begin{aligned}\psi(O;\tau_0,\eta_\varepsilon) &= \big(Z - \pi_0(W) - \varepsilon h_\pi(W)\big)\Big(U_0(O) - \varepsilon(h_\mu(W) - \tau_0 h_r(W))\Big)\\ &= \underbrace{(Z - \pi_0(W))U_0(O)}_{\psi(O;\tau_0,\eta_0)} - \varepsilon\, h_\pi(W)U_0(O) - \varepsilon\,(Z - \pi_0(W))(h_\mu(W) - \tau_0 h_r(W))\\ &\qquad + \varepsilon^2\, h_\pi(W)\,(h_\mu(W) - \tau_0 h_r(W)).\end{aligned}$$

Step 2: Taking expectations and differentiating at $\varepsilon = 0$ (componentwise) yields

$$D_\eta M(\tau_0,\eta_0)[h] = \frac{d}{d\varepsilon}E[\psi(O;\tau_0,\eta_\varepsilon)]\Big|_{\varepsilon=0} = -E[h_\pi(W)U_0(O)] - E[(Z - \pi_0(W))(h_\mu(W) - \tau_0 h_r(W))]\,.$$

Step 3: For the first term, note that $\mu_0(W) = E[Y \mid W]$ and $r_0(W) = E[D \mid W]$, so

$$E[U_0(O) \mid W] = E[Y - \mu_0(W) \mid W] - \tau_0 E[D - r_0(W) \mid W] = 0.$$

Hence

$$E[h_\pi(W)U_0(O)] = E[h_\pi(W)\,E[U_0(O) \mid W]] = 0.$$

For the second term, $\pi_0(W) = E[Z \mid W]$ implies $E[Z - \pi_0(W) \mid W] = 0$, so

$$E[(Z - \pi_0(W))(h_\mu(W) - \tau_0 h_r(W))] = E[E[Z - \pi_0(W) \mid W]\,(h_\mu(W) - \tau_0 h_r(W))] = 0.$$

Combining these equalities gives $D_\eta M(\tau_0,\eta_0)[h] = 0 \in \mathbb{R}^{d_z}$ for all directions h.

B.3. Proof of Lemma 2 (Second-order moment decomposition)

Fix $\tau = \tau_0$ and recall the score

$$\psi(O;\tau_0,\eta) = (Z - \pi(W))\Big((Y - \mu(W)) - \tau_0(D - r(W))\Big), \qquad M(\tau_0,\eta) = E[\psi(O;\tau_0,\eta)].$$

Define

$$m(W) := \mu(W) - \tau_0 r(W), \qquad m_0(W) := \mu_0(W) - \tau_0 r_0(W) = E[Y - \tau_0 D \mid W].$$

tep 1: Observe that

$$(Y - \mu(W)) - \tau_0(D - r(W)) = (Y - \tau_0 D) - (\mu(W) - \tau_0 r(W)) = (Y - \tau_0 D) - m(W).$$

Therefore

$$\psi(O;\tau_0,\eta) = (Z - \pi(W))\big((Y - \tau_0 D) - m(W)\big).$$

Step 2: Write

$$(Y - \tau_0 D) - m(W) = \big((Y - \tau_0 D) - m_0(W)\big) + \big(m_0(W) - m(W)\big).$$

Hence

$$\psi(O;\tau_0,\eta) = (Z - \pi(W))\big((Y - \tau_0 D) - m_0(W)\big) + (Z - \pi(W))\big(m_0(W) - m(W)\big).$$

Taking expectations,

$$M(\tau_0,\eta) = E[(Z - \pi(W))((Y - \tau_0 D) - m_0(W))] + E[(Z - \pi(W))(m_0(W) - m(W))]. \quad \text{(B.1)}$$

Step 3: From the PLIV model $Y = \tau_0 D + g_0(W) + \zeta$ with $E[\zeta \mid Z, W] = 0$, I have

$$Y - \tau_0 D = g_0(W) + \zeta, \qquad m_0(W) = E[Y - \tau_0 D \mid W] = g_0(W) + E[\zeta \mid W] = g_0(W),$$

since $E[\zeta \mid W] = E[E[\zeta \mid Z, W] \mid W] = 0$. Therefore

$$(Y - \tau_0 D) - m_0(W) = \zeta.$$

Because $(Z - \pi(W))$ is measurable with respect to (Z, W), Assumption 1 gives

$$E[(Z - \pi(W))((Y - \tau_0 D) - m_0(W))] = E[(Z - \pi(W))\zeta] = E[(Z - \pi(W))E[\zeta \mid Z, W]] = 0.$$

Step 4: Since $(m_0(W) - m(W))$ is a function of W only, apply iterated expectations to the second term in (B.1):

$$\begin{aligned} E[(Z - \pi(W))(m_0(W) - m(W))] &= E[E[Z - \pi(W) \mid W]\,(m_0(W) - m(W))] \\ &= E[(E[Z \mid W] - \pi(W))(m_0(W) - m(W))] \\ &= E[(\pi_0(W) - \pi(W))(m_0(W) - m(W))]. \end{aligned}$$

This proves the exact identity (3.4).

Step 5: By Cauchy–Schwarz,

$$\|M(\tau_0, \eta)\|_2 = \|E[(\pi_0 - \pi)(m_0 - m)]\|_2 \le \|\pi - \pi_0\|_{P,2}\, \|m - m_0\|_{P,2}.$$

Moreover, since $m - m_0 = (\mu - \mu_0) - \tau_0(r - r_0)$, the triangle inequality yields

$$\|m - m_0\|_{P,2} \le \|\mu - \mu_0\|_{P,2} + |\tau_0|\, \|r - r_0\|_{P,2}.$$

Combining these inequalities gives the bound in the lemma. Finally, $M(\tau_0, \eta) = 0$ whenever either (i) $\pi = \pi_0$ a.s. or (ii) $(\mu, r) = (\mu_0, r_0)$ a.s., because then at least one factor in $E[(\pi_0 - \pi)(m_0 - m)]$ is identically zero.

B.4. Proof of Lemma 3 (Foldwise second-order bound)

Fix an outer fold $k \in \{1, \dots, K\}$. By construction of the nested-CV protocol in Section 2.4, the nuisance estimator

$$\hat{\eta}^{(-k)} = (\hat{\pi}^{(-k)}, \hat{\mu}^{(-k)}, \hat{r}^{(-k)})$$

is computed using only the training sample $\mathcal{I}_{-k}$ (including any inner/nested CV), and therefore is measurable with respect to the σ-field generated by the training data.

Let $O = (Y, D, Z, W)$ denote a generic draw from the population distribution, independent of the training sample. Conditional on $\mathcal{I}_{-k}$, the functions $\hat{\pi}^{(-k)}, \hat{\mu}^{(-k)}, \hat{r}^{(-k)}$ are fixed, so I can take conditional expectations holding these functions fixed.

Step 1: Recall $M(\tau, \eta) = E[\psi(O; \tau, \eta)]$ and that $M(\tau_0, \eta_0) = 0$ by identification. Then

$$\begin{aligned} E\Big[\psi(O; \tau_0, \hat{\eta}^{(-k)}) - \psi(O; \tau_0, \eta_0) \mid \mathcal{I}_{-k}\Big] &= E\Big[\psi(O; \tau_0, \hat{\eta}^{(-k)}) \mid \mathcal{I}_{-k}\Big] - E[\psi(O; \tau_0, \eta_0)] \\ &= M(\tau_0, \hat{\eta}^{(-k)}) - M(\tau_0, \eta_0) = M(\tau_0, \hat{\eta}^{(-k)}). \end{aligned}$$

Step 2: Define

$$\hat{m}^{(-k)}(W) := \hat{\mu}^{(-k)}(W) - \tau_0 \hat{r}^{(-k)}(W).$$

By Lemma 2, applied conditional on $\mathcal{I}_{-k}$ (treating $\hat{\eta}^{(-k)}$ as fixed),

$$M(\tau_0, \hat{\eta}^{(-k)}) = E\Big[(\pi_0(W) - \hat{\pi}^{(-k)}(W))\,(m_0(W) - \hat{m}^{(-k)}(W))\Big] \in \mathbb{R}^{d_z}.$$

Step 3: Using Cauchy–Schwarz for vector-valued expectations,

$$\left\| M(\tau_0, \hat{\eta}^{(-k)}) \right\|_2 \leq \|\hat{\pi}^{(-k)} - \pi_0\|_{P,2} \, \|\hat{m}^{(-k)} - m_0\|_{P,2}.$$

Moreover,

$$\hat{m}^{(-k)}(W) - m_0(W) = (\hat{\mu}^{(-k)}(W) - \mu_0(W)) - \tau_0(\hat{r}^{(-k)}(W) - r_0(W)),$$

so by the triangle inequality,

$$\|\hat{m}^{(-k)} - m_0\|_{P,2} \leq \|\hat{\mu}^{(-k)} - \mu_0\|_{P,2} + |\tau_0| \, \|\hat{r}^{(-k)} - r_0\|_{P,2}.$$

Combining the last two displays yields exactly the claimed bound:

$$\left\| E\left[\psi(O; \tau_0, \hat{\eta}^{(-k)}) - \psi(O; \tau_0, \eta_0) \mid \mathcal{I}_{-k}\right] \right\|_2 \leq \|\hat{\pi}^{(-k)} - \pi_0\|_{P,2} \Big(\|\hat{\mu}^{(-k)} - \mu_0\|_{P,2} + |\tau_0| \, \|\hat{r}^{(-k)} - r_0\|_{P,2} \Big).$$

Step 4: If $\|\hat{\pi}^{(-k)} - \pi_0\|_{P,2} = o_p(n^{-1/4})$ and $\|\hat{\mu}^{(-k)} - \mu_0\|_{P,2} + \|\hat{r}^{(-k)} - r_0\|_{P,2} = o_p(n^{-1/4})$ uniformly in k, then the right-hand side is $o_p(n^{-1/2})$ uniformly in k. This is the stated foldwise bias order.

C. ALGORITHM

Algorithm 1 makes the key measurability explicit: conditional on the training sample $\mathcal{I}_{-k}$, the selected learner and fitted nuisance functions are fixed, and the evaluation fold $\mathcal{I}_k$ is not used for selection. This is the precise sense in which nested CV prevents leakage and preserves standard DML expansions.

D. ADDITIONAL MONTE CARLO DETAILS

This appendix reports the full Monte Carlo design and the complete simulation results underlying the summary in Section 4. All results are based on 100 Monte Carlo replications, $K = 3$ outer cross-fitting folds, and $J = 3$ inner folds.

D.1. Data-generating processes and implementation details

For each replication, I generate i.i.d. observations $\{(Y_i, D_i, Z_i, W_i)\}_{i=1}^n$. The controls are high-dimensional Gaussian covariates,

$$W_i \sim \mathcal{N}(0, \Sigma), \qquad \Sigma_{jk} = 0.5^{|j-k|},$$

and the instrument is vector-valued with fixed dimension $d_z = 5$,

$$Z_i = (Z_{i1}, \ldots, Z_{i5})', \qquad Z_{ij} \overset{i.i.d.}{\sim} \mathcal{U}(0, 1),$$

independently of W_i. Endogeneity is induced by an unobserved confounder $A_i \sim \mathcal{N}(0, 1)$, independent of (Z_i, W_i), and the data are generated as

$$D_i = m(Z_i, W_i) + A_i + \nu_i, \tag{D.2}$$

$$Y_i = \tau_0 D_i + g(W_i) + A_i + \varepsilon_i, \tag{D.3}$$

where $\tau_0 = 1$ and $\nu_i, \varepsilon_i \overset{i.i.d.}{\sim} \mathcal{N}(0, 1)$.

Two nuisance designs are considered.

Algorithm 1 Nested-CV DML-IV Estimation Protocol (Residualized 2SLS)

Input: Data $\{(Y_i, D_i, Z_i, W_i)\}_{i=1}^n$, learner library $\mathcal{L}$ (learner classes and tuning-parameter grids), outer folds K, inner folds J.
Partition $\{1, \dots, n\}$ into K disjoint folds $\mathcal{I}_1, \dots, \mathcal{I}_K$.
for $k = 1, \dots, K$ **do**
  Define training indices $\mathcal{I}_{-k} = \{1, \dots, n\} \setminus \mathcal{I}_k$ and evaluation indices $\mathcal{I}_k$.
  Partition $\mathcal{I}_{-k}$ into J inner folds $\mathcal{I}_{-k,1}, \dots, \mathcal{I}_{-k,J}$.
  (Inner-CV criterion) For a target T and regressors X, define

$$CV_J(T \mid X; \ell) := \frac{1}{J} \sum_{m=1}^{J} \frac{1}{|\mathcal{I}_{-k,m}|} \sum_{i \in \mathcal{I}_{-k,m}} \big(T_i - \hat{f}_\ell^{(-k,-m)}(X_i)\big)^2,$$

  where $\hat{f}_\ell^{(-k,-m)}$ is fit using learner ℓ on $\mathcal{I}_{-k} \setminus \mathcal{I}_{-k,m}$.
  (Tuning) If learner class ℓ has a tuning grid Λ_ℓ, interpret $CV_J(T \mid X; \ell) := \min_{\lambda \in \Lambda_\ell} CV_J(T \mid X; (\ell, \lambda))$ and let $\hat{\lambda}_\ell$ denote the minimizer.
  Select/fit $\hat{\mu}^{(-k)}$**:** $\hat{\ell}_Y^* = \arg\min_{\ell \in \mathcal{L}} CV_J(Y \mid W; \ell)$; refit $\hat{\mu}^{(-k)}$ on all $\mathcal{I}_{-k}$ using $\hat{\ell}_Y^*$.
  Select/fit $\hat{r}^{(-k)}$**:** $\hat{\ell}_D^* = \arg\min_{\ell \in \mathcal{L}} CV_J(D \mid W; \ell)$; refit $\hat{r}^{(-k)}$ on all $\mathcal{I}_{-k}$ using $\hat{\ell}_D^*$.
  Select/fit $\hat{\pi}^{(-k)}$**:** for each $j = 1, \dots, d_z$, set $\hat{\ell}_{Z,j}^* = \arg\min_{\ell \in \mathcal{L}} CV_J(Z_j \mid W; \ell)$; refit $\hat{\pi}_j^{(-k)}$ on all $\mathcal{I}_{-k}$; set $\hat{\pi}^{(-k)} = (\hat{\pi}_1^{(-k)}, \dots, \hat{\pi}_{d_z}^{(-k)})'$.
  for $i \in \mathcal{I}_k$ **do**
    $\tilde{Y}_i = Y_i - \hat{\mu}^{(-k)}(W_i), \;\; \tilde{D}_i = D_i - \hat{r}^{(-k)}(W_i), \;\; \tilde{Z}_i = Z_i - \hat{\pi}^{(-k)}(W_i).$
  end for
  End fold k **(store** $\{(\tilde{Y}_i, \tilde{D}_i, \tilde{Z}_i) : i \in \mathcal{I}_k\}$**).**
end for
End outer-fold loop.
Form cross-fitted moments: $\hat{\Omega} = \frac{1}{n} \sum_{i=1}^n \tilde{Z}_i \tilde{Z}_i', \;\; \hat{g}_{ZD} = \frac{1}{n} \sum_{i=1}^n \tilde{Z}_i \tilde{D}_i, \;\; \hat{g}_{ZY} = \frac{1}{n} \sum_{i=1}^n \tilde{Z}_i \tilde{Y}_i.$
Compute $\hat{\Omega}^{-1}$ (or a generalized inverse if $\hat{\Omega}$ is singular).
Compute the residualized 2SLS/GMM estimator:

$$\hat{\tau} = \frac{\hat{g}_{ZD}' \hat{\Omega}^{-1} \hat{g}_{ZY}}{\hat{g}_{ZD}' \hat{\Omega}^{-1} \hat{g}_{ZD}}.$$

Compute score residuals $\hat{u}_i := \tilde{Y}_i - \hat{\tau}\, \tilde{D}_i$ and moments $\hat{\psi}_i := \tilde{Z}_i \hat{u}_i$, $\hat{\Sigma} := \frac{1}{n} \sum_{i=1}^n \hat{\psi}_i \hat{\psi}_i'$.
Compute the plug-in asymptotic variance

$$\hat{V} = \frac{\hat{g}_{ZD}' \hat{\Omega}^{-1} \hat{\Sigma}\, \hat{\Omega}^{-1} \hat{g}_{ZD}}{\big(\hat{g}_{ZD}' \hat{\Omega}^{-1} \hat{g}_{ZD}\big)^2}, \qquad \widehat{\mathrm{se}}(\hat{\tau}) = \sqrt{\hat{V}/n}.$$

If interval inference is desired, form $\hat{\tau} \pm z_{1-\alpha/2}\, \widehat{\mathrm{se}}(\hat{\tau})$.

Sparse design.

$$m(Z_i, W_i) = 0.5 \sum_{j=1}^{5} Z_{ij} + 0.5 \sum_{k=1}^{5} W_{ik}, \qquad g(W_i) = 0.5 \sum_{k=1}^{5} W_{ik}.$$

Nonlinear design.

$$m(Z_i, W_i) = Z_{i1} Z_{i2} + \sin(W_{i1}) \cos(W_{i2}) + \exp(0.5 W_{i3}),$$

$$g(W_i) = \frac{1}{1 + e^{-W_{i1}}} + (W_{i2} W_{i3}) + \mathbb{I}(W_{i4} > 0).$$

I report results for $n \in \{1000, 3000, 10000\}$. Each simulated dataset is estimated using the leakage-safe nested-CV protocol described in Algorithm 1. Within each outer training fold, nui-

sance learners and tuning parameters are selected by inner cross-validation using training observations only, refit on the full outer training sample, and evaluated on the held-out fold. For the instrument regression $\pi_0(W) = E[Z \mid W]$ with $d_z > 1$, estimation is implemented coordinate-wise.

For each learner and each design, I report the Monte Carlo mean of $\hat{\tau}$, its bias relative to $\tau_0 = 1$, the Monte Carlo standard deviation of $\hat{\tau}$, the root mean squared error (RMSE), the average estimated standard error, and empirical coverage of nominal 95% confidence intervals. Tables 1–2 reinforce the main-text summary. In the sparse design, Nested-CV is effectively tied with the strongest penalized linear learners and remains competitive with the better nonlinear baselines. In the nonlinear design, Nested-CV improves materially on fixed linear specifications and performs comparably to strong nonlinear learners, while avoiding the pronounced instability of the MLP specification in finite samples.

Table 1: Full Monte Carlo results: mean estimate, average standard error, and coverage

	Mean($\hat{\tau}$)			Avg(SE)			Coverage (%)		
Model	$n = 1000$	$n = 3000$	$n = 10000$	$n = 1000$	$n = 3000$	$n = 10000$	$n = 1000$	$n = 3000$	$n = 10000$
Panel A: Sparse design									
Adaptive Nested-CV	1.032	0.996	1.008	0.138	0.081	0.044	93	98	96
Ridge	1.045	0.997	1.009	0.135	0.080	0.043	83	95	95
Lasso	1.032	0.997	1.008	0.138	0.082	0.044	93	96	96
Elastic Net	1.032	0.996	1.007	0.139	0.081	0.044	93	98	95
Random Forest	1.022	1.000	1.009	0.156	0.089	0.047	95	97	94
Gradient Boosting	1.029	0.995	1.009	0.159	0.088	0.045	94	95	96
XGBoost	1.025	1.000	1.010	0.160	0.088	0.045	94	96	96
MLP	0.994	0.933	1.202	0.108	0.064	0.039	83	60	6
Panel B: Nonlinear design									
Adaptive Nested-CV	1.084	1.021	1.012	0.255	0.142	0.074	90	98	94
Ridge	1.125	1.037	1.013	0.266	0.159	0.089	83	97	93
Lasso	1.091	1.031	1.014	0.273	0.162	0.089	91	98	92
Elastic Net	1.096	1.032	1.015	0.275	0.161	0.089	92	98	92
Random Forest	1.081	1.024	1.016	0.254	0.142	0.076	89	95	93
Gradient Boosting	1.073	1.035	1.016	0.270	0.142	0.074	92	96	90
XGBoost	1.068	1.035	1.012	0.269	0.142	0.074	93	95	92
MLP	1.098	1.067	1.011	0.104	0.077	0.063	80	84	80

Notes: Entries report Monte Carlo means of the PLIV estimator, average estimated standard errors, and empirical coverage of nominal 95% confidence intervals over 100 replications.

E. SUPPLEMENTARY OUTPUT FOR COLLEGEDISTANCE

This appendix reports two diagnostics for the leakage-safe Nested-CV implementation used in the PLIV DML-IV exercise. The goal is not to advance a structural interpretation of the point estimates, but to document (i) the *effective* instrument strength after cross-fitted residualization and (ii) the stability of the resulting Nested-CV estimates across random sample splits.

Two patterns are worth noting for transparency Tables 3–4. First, the PLIV DML-IV estimates can differ nontrivially across nuisance learner classes in this application: linear learners deliver estimates close to the benchmark 2SLS column, while more flexible learners (e.g., tree-based methods) produce estimates closer to zero under the same orthogonal score. Second, the Nested-CV protocol selects learners purely by *out-of-fold predictive performance* within each training split, and the selection frequencies in Table 4 indicate that flexible learners are often preferred for $\mu_0(W)$ and (under the full library) for $\pi_0(W)$. Note that these supplementary outputs are included as sensitivity and transparency checks for the estimation pipeline, not as a basis for structural interpretation of the resulting point estimates.

Table 2: Full Monte Carlo results: bias, standard deviation, and RMSE

	Bias($\hat{\tau}$)			SD($\hat{\tau}$)			RMSE($\hat{\tau}$)		
Model	$n = 1000$	$n = 3000$	$n = 10000$	$n = 1000$	$n = 3000$	$n = 10000$	$n = 1000$	$n = 3000$	$n = 10000$
Panel A: Sparse design									
Adaptive Nested-CV	0.032	-0.004	0.008	0.135	0.084	0.039	0.138	0.084	0.039
Ridge	0.045	-0.003	0.009	0.166	0.089	0.039	0.171	0.089	0.040
Lasso	0.032	-0.003	0.008	0.134	0.086	0.038	0.137	0.085	0.039
Elastic Net	0.032	-0.004	0.007	0.136	0.084	0.039	0.138	0.083	0.039
Random Forest	0.022	0.000	0.009	0.147	0.091	0.042	0.148	0.090	0.043
Gradient Boosting	0.029	-0.005	0.009	0.149	0.092	0.041	0.151	0.092	0.042
XGBoost	0.025	-0.000	0.010	0.164	0.092	0.042	0.165	0.092	0.043
MLP	-0.006	-0.067	0.202	0.143	0.115	0.097	0.143	0.133	0.224
Panel B: Nonlinear design									
Adaptive Nested-CV	0.084	0.021	0.012	0.258	0.121	0.073	0.270	0.123	0.073
Ridge	0.125	0.037	0.013	0.362	0.151	0.093	0.381	0.155	0.094
Lasso	0.091	0.031	0.014	0.289	0.141	0.092	0.302	0.144	0.092
Elastic Net	0.096	0.032	0.015	0.293	0.141	0.091	0.307	0.144	0.091
Random Forest	0.081	0.024	0.016	0.267	0.125	0.074	0.277	0.127	0.075
Gradient Boosting	0.073	0.035	0.016	0.285	0.131	0.073	0.293	0.135	0.075
XGBoost	0.068	0.035	0.012	0.293	0.136	0.074	0.300	0.139	0.075
MLP	0.098	0.067	0.011	0.111	0.087	0.102	0.148	0.110	0.102

Notes: Bias is computed relative to the true parameter $\tau_0 = 1$. SD denotes the Monte Carlo standard deviation of $\hat{\tau}$, and RMSE denotes the root mean squared error over 100 replications.

Table 3: CollegeDistance: PLIV DML-IV estimates across learner classes (full output)

	Baseline controls (W_{base}, $p = 11$)		High-dimensional controls (W_{hd}, $p = 77$	
Estimator	$\hat{\tau}$	SE	$\hat{\tau}$	SE
OLS (benchmark)	-0.0009	0.0011	-0.0009	0.0011
2SLS (benchmark)	0.0519	0.0200	0.0519	0.0200
Fixed-learner DML-IV (cross-fitted)				
DML (Linear Regression)	0.0536	0.0197	0.0705	0.0229
DML (Ridge)	0.0537	0.0197	0.0693	0.0226
DML (Lasso)	0.0531	0.0198	0.0588	0.0212
DML (Elastic Net)	0.0536	0.0198	0.0660	0.0219
DML (Random Forest)	0.0015	0.0016	0.0026	0.0021
DML (Gradient Boosting)	-0.0084	0.0051	0.0060	0.0053
DML (XGBoost)	-0.0066	0.0068	0.0045	0.0054
DML (MLP)	0.0394	0.0207	0.0272	0.0246
Leakage-safe protocol				
Nested-CV (full π library)	0.0016	0.0018	0.0029	0.0024
Nested-CV (restricted π library)	-0.0003	0.0004	0.0013	0.0012

Notes: $n = 4{,}739$. Outcome is log wage; treatment is years of education; instrument is distance to a four-year college. DML-IV us $K = 5$ outer-fold cross-fitting and leakage-free nested CV with $K_{\text{inner}} = 3$ folds inside each training split. Standard errors are bas on the orthogonal PLIV score. 95% confidence intervals are $\hat{\tau} \pm 1.96 \cdot \text{SE}$. The restricted π library limits the instrument regressi $\pi_0(W) = \mathbb{E}[Z \mid W]$ to OLS/Ridge/Lasso/Elastic Net.

Table 4: CollegeDistance: Nested-CV selection frequencies across outer folds

Controls	Variant	$\mu_0(W)$	$r_0(W)$	$\pi_0(W)$
W_{base}	full π lib	RF (5/5)	XGBoost (2/5), GB (2/5), Lasso (1/5)	RF (5/5)
W_{base}	restricted π lib	RF (5/5)	XGBoost (2/5), GB (2/5), Lasso (1/5)	Lasso (2/5), Ridge (2/5), EN (1/5)
W_{hd}	full π lib	RF (5/5)	EN (2/5), XGBoost (1/5), GB (1/5), Lasso (1/5)	RF (5/5)
W_{hd}	restricted π lib	RF (5/5)	EN (2/5), XGBoost (1/5), GB (1/5), Lasso (1/5)	Ridge (2/5), OLS (1/5), EN (1/5), Lasso (1/5)

Notes: Entries report the learner(s) selected by *inner-CV MSE* within each outer training fold, along with selection counts out of $K = 5$. Selections are reported separately for the outcome regression $\mu_0(W) = \mathbb{E}[Y \mid W]$, treatment regression $r_0(W) = \mathbb{E}[D \mid W]$, and instrument regression $\pi_0(W) = \mathbb{E}[Z \mid W]$. RF denotes Random Forest; GB denotes Gradient Boosting; EN denotes Elastic Net. These frequencies summarize the *predictive* choices made by the Nested CV protocol and are not interpreted as evidence about the causal parameter.

The first diagnostic is a "residualized first stage" computed using the cross-fitted residuals $\tilde{D} = D - \hat{r}(W)$ and $\tilde{Z} = Z - \hat{\pi}(W)$ produced by the Nested-CV procedure. The F-statistic and R^2 are reported from an OLS regression of $\tilde{D}$ on $\tilde{Z}$ (including a constant), which summarizes instrument strength *after* partialling out controls via cross-fitted nuisance estimates. The second diagnostic re-runs the entire Nested-CV procedure over 20 different outer-fold random seeds and summarizes the distribution of the resulting $\hat{\tau}$ estimates and their reported standard errors.

Table 5: CollegeDistance: residualized first-stage diagnostics (Nested-CV, high-dimensional controls)

Variant	Residualized F-statistic	Residualized R^2
Full π library	10.64	0.0022
Restricted π library	18.67	0.0039

Notes: Computed from cross-fitted residuals $\tilde{D} = D - \hat{r}(W)$ and $\tilde{Z} = Z - \hat{\pi}(W)$ under the Nested-CV procedure (high-dimensional controls). The residualized F and R^2 come from the auxiliary regression $\tilde{D}$ on $\tilde{Z}$ with an intercept.

Taken together, Tables 5–6 indicate that the *residualized* first-stage signal is modest in explained variation (residualized R^2 well below 1%), and that the corresponding Nested-CV PLIV estimates are tightly concentrated near zero across random splits. This combination is consistent with a setting in which, under the chosen control specifications and nuisance learners, the cross-fitted partialling-out step leaves limited residualized instrument variation available for identifying the target moment condition.

F. WEAK-IV STRESS TEST: ANGRIST–KRUEGER QUARTER-OF-BIRTH

This appendix reports an additional benchmark based on the quarter-of-birth design of Angrist and Krueger (1991). The purpose is not to resolve weak-IV identification debates, but to stress-test the proposed nuisance-selection protocol in a setting where identifying variation is plausibly limited. Accordingly, these results are interpreted as a robustness exercise rather than as the headline empirical application. (Concerns about weak instruments in IV applications motivate careful inference in such settings; see, e.g., Staiger and Stock, 1997; Hansen, Hausman, and Newey, 2008.)

Table 6: CollegeDistance: repeat-split stability of Nested-CV estimates (20 random seeds)

Variant	Mean($\hat{\tau}$)	SD($\hat{\tau}$)	Mean(SE)
Full π library	0.0021	0.0020	0.0033
Restricted π library	0.0002	0.0010	0.0013

Notes: Each row summarizes 20 reruns of the Nested-CV estimator using different random seeds for the outer-fold split.

Table 7: CollegeDistance baseline first-stage diagnostics

N	d_z	First-stage F	p-value	Partial R^2
4,739	1	22.252	< 0.001	0.0047

Notes: Diagnostics are computed from the raw first-stage regression $D \sim Z + W$ using the baseline CollegeDistance control set.

For comparison, the main-text CollegeDistance application features a stronger excluded instrument than the quarter-of-birth design used here. Under baseline controls, the CollegeDistance first-stage F statistic is 22.25 with partial $R^2 = 0.0047$ (Table 7), while the quarter-of-birth appendix reports substantially weaker first-stage diagnostics in Table 9. This appendix is therefore best interpreted as a stress test of the workflow under limited identifying variation, not as a setting in which Wald-style inference is expected to be robust to weak identification.

F.1. Specification and implementation

I estimate the causal effect of schooling on log wages using an IV strategy based on quarter-of-birth interactions with cohort indicators, controlling for a rich set of observed covariates as in Angrist and Krueger (1991). Estimation follows Algorithm 1. In each outer cross-fitting fold, the nuisance regressions $\mu_0(W) = E[Y \mid W]$, $r_0(W) = E[D \mid W]$, and $\pi_0(W) = E[Z \mid W]$ are estimated using nested cross-validation within the training sample only, selecting among penalized linear learners and nonlinear learners. The orthogonal score is then evaluated on the held-out fold, and $\hat{\tau}$ is computed from the corresponding projected GMM / residualized 2SLS first-order condition, with the plug-in standard error reported in Section 3.5.

F.2. Results

Table 8 reports OLS, TSLS, fixed-learner DML-IV estimates, and the leakage-safe Nested-CV estimate. Across learner classes, the resulting standard errors and 95% confidence intervals are broadly comparable in this application. The main practical value of the nested-CV protocol is therefore not a mechanical tightening of inference, but a disciplined and reproducible way to select and tune nuisance learners strictly within training folds while preserving the orthogonal-score cross-fitting logic.

Three patterns stand out. First, the plug-in standard errors are very similar across learner classes in this application. Second, the fixed-learner DML estimates remain close to the TSLS benchmark, suggesting limited sensitivity to nuisance choice in point estimates relative to the stronger functional-form sensitivity observed in the CollegeDistance application. Third, the main empirical lesson here is procedural rather than inferential: the leakage-safe protocol permits

Table 8: Returns to Education: Fixed Learners vs. Nested-CV Protocol

Estimation strategy	$\hat{\tau}$	**SE**	**95% CI**
Panel A: Classical baselines			
OLS	0.0694	(0.0004)	(0.0686, 0.0701)
TSLS	0.0983	(0.0137)	(0.0714, 0.1252)
Panel B: Fixed-learner DML-IV (cross-fitted)			
DML (Linear Regression)	0.0983	(0.0142)	(0.0705, 0.1261)
DML (Ridge)	0.0983	(0.0142)	(0.0705, 0.1261)
DML (Lasso)	0.0984	(0.0142)	(0.0706, 0.1262)
DML (Elastic Net)	0.0984	(0.0142)	(0.0706, 0.1262)
DML (Random Forest)	0.0974	(0.0142)	(0.0696, 0.1253)
DML (Gradient Boosting)	0.0958	(0.0140)	(0.0683, 0.1233)
DML (XGBoost)	0.0952	(0.0140)	(0.0678, 0.1225)
Panel C: Leakage-safe protocol			
DML (Nested-CV)	**0.0982**	**(0.0142)**	**(0.0705, 0.1260)**

Notes: DML-IV uses the residualized PLIV orthogonal score and $K = 5$ outer-fold cross-fitting. Fixed-learner rows use the same learner class for each nuisance regression, with tuning performed by inner cross-validation on training folds only.

disciplined adaptive tuning even in a weak-first-stage environment, while the reported Wald intervals should not be interpreted as weak-IV-robust confidence sets.

Table 9 reports a modest first-stage F statistic and a small partial R^2, reflecting the well-known weak first-stage properties of quarter-of-birth instruments in this setting. The asymptotic normality result in Section 3.5 is developed under standard *non-weak* identification conditions. Accordingly, the Wald-style confidence intervals reported in Table 8 should be interpreted primarily as an illustration of the leakage-safe nested-CV DML-IV workflow and its stability properties, rather than as weak-IV-robust inference.

Table 9: First-stage diagnostics for excluded quarter-of-birth interaction instruments.

N	d_z	First-stage F	p-value	Partial R^2
329,509	54	3.848	< 0.001	0.0006

Note: N represents the sample size and d_z is the number of excluded instruments. The joint F-statistic, p-value, and partial R^2 correspond to the excluded quarter-of-birth interaction instruments in the reduced-form first-stage regression.